

Program : Diploma in Instrumentation Engineering	
Course Code : 5339A	Course Title: Industrial Robotics Lab
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To study the position of links.
- To study the transformation matrix usage.
- To study the robot application for pick and place.
- To understand Trajectory Control.
- To calculate the joint torque of a robot.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Basic Knowledge in robotics		Foundations of robotics, Robotic case Studies	2,3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Perform modelling of robot based on the link positions and Determine the orientation and position of robot.	9	Applying
CO2	Develop pick and place application system.	12	Applying
CO3	Integrate PLC and Robot.	9	Applying
CO4	Trajectory Control Modelling and Joint Torques of Robot	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2	3						
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Perform modelling of robot based on the link positions and Determine the orientation and position of robot.		
M1.01	Determination of maximum and minimum position of links.	3	Applying
M1.02	Verification of transformation (Position and orientation) with respect to gripper and world coordinate system.	3	Applying
M1.03	Estimation of accuracy, repeatability and resolution.	3	Applying
CO2	Develop pick and place application system.		
M2.01	Control the Velocity of the Joints.	6	Applying
M2.02	Robot programming and simulation for pick and place.	6	Applying
	Lab Test – II	2	
CO3	Integrate PLC and Robot.		
M3.01	Robot programming and simulation for Colour identification	3	Applying
M3.02	Robot programming and simulation for Shape identification	3	Applying
M3.03	Robot programming and simulation for assembly process	3	Applying
CO4	Trajectory Control Modelling and Joint Torques of Robot		

M4.01	Trajectory Control Modelling with Inverse Kinematics	6	Applying
M4.02	Robot programming for joint torque calculation	6	Applying
	Lab Test – II	1	

Text / Reference

T/R	Book Title/Author
T1	Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and A. K. Deb, Jaico Publishing House, 2013
R3	Krishnakanth, Computer based Industrial Control, PHI.
R4	Donald .P. Eckman, Automatic process control, WILEY India.